In [77]:
```
import pandas as pd
```

In [78]:
```
# Load the messy hospital dataset from Excel
df = pd.read_excel("/Users/naoko/Desktop/messy_hospital_data.xlsx")

# Display the dataset to understand its structure and contents
df
```

Out[78]:

	Date of Admission	Patient Name	Gender	Age	Blood Type	Region	Medical Condition	Outcon
0	2024-01-12	KriStOpHeRlOPEZ	Female	47	AB	Illinois	Cancer	Succe
1	2024-08-02	JenNIferBRowN	Female	68	B-	Illinois	Flu	Failu
2	2023-03-06	JamiEluCAS	Female	46	O	California	Infection	Succe
3	2024-11-09	wayneBROwn	Male	88	B-	Illinois	Injury	Failu
4	2023-12-20	JeNnIfeRfullEr	Male	68	A+	Texas	Hypertension	Failu
...	...	...	...	...	...	...	...	
1995	2024-02-02	EdwaRDrOMAn	Male	29	AB	Florida	Diabetes	Succe
1996	2023-06-16	bARBaRasHaH DVm	Male	65	B-	Illinois	Flu	Failu
1997	2022-08-09	reBeccadAvIS	Male	10	B	New York	Flu	Succe
1998	2022-01-02	StEpHENdAniElS	Female	59	B	Illinois	Injury	Succe
1999	2021-08-30	AAroNlOgaN	Female	49	A	Texas	Hypertension	Succe

2000 rows × 9 columns

In [79]:
```
# Remove duplicate rows from the dataset
df = df.drop_duplicates()

# Review the updated dataset
df
```

Out[79]:

	Date of Admission	Patient Name	Gender	Age	Blood Type	Region	Medical Condition	Outcon
0	2024-01-12	KriStOpHeRlOPEZ	Female	47	AB	Illinois	Cancer	Succe
1	2024-08-02	JenNIferBRowN	Female	68	B-	Illinois	Flu	Failu
2	2023-03-06	JamiEluCAS	Female	46	O	California	Infection	Succe
3	2024-11-09	wayneBROwn	Male	88	B-	Illinois	Injury	Failu
4	2023-12-20	JeNnIfeRfullEr	Male	68	A+	Texas	Hypertension	Failu
...	...	...	...	...	...	...	...	
1995	2024-02-02	EdwaRDrOMAn	Male	29	AB	Florida	Diabetes	Succe
1996	2023-06-16	bARBaRasHaH DVm	Male	65	B-	Illinois	Flu	Failu
1997	2022-08-09	reBeccadAvIS	Male	10	B	New York	Flu	Succe
1998	2022-01-02	StEpHENdAniEIS	Female	59	B	Illinois	Injury	Succe
1999	2021-08-30	AAroNlOgaN	Female	49	A	Texas	Hypertension	Succe

2000 rows × 9 columns

In [80]:

```
# Check the data type of the Date of Admission column
df["Date of Admission"].dtype

# Check overall data types and non-null counts
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Date of Admission  2000 non-null   datetime64[us]
 1   Patient Name       2000 non-null   str
 2   Gender             2000 non-null   str
 3   Age                2000 non-null   int64
 4   Blood Type         2000 non-null   str
 5   Region             2000 non-null   str
 6   Medical Condition  2000 non-null   str
 7   Outcome            2000 non-null   str
 8   Date of Discharge  2000 non-null   datetime64[us]
dtypes: datetime64[us](2), int64(1), str(6)
memory usage: 140.8 KB
```

In [81]:

```
# Standardize name format (e.g., "JenNIfer BRowN" → "Jennifer Brown")
df["Patient Name"] = df["Patient Name"].str.strip().str.title()
```

```
# Review the updated dataset
df
```

Out[81]:

	Date of Admission	Patient Name	Gender	Age	Blood Type	Region	Medical Condition	Outcome
0	2024-01-12	Kristopher Lopez	Female	47	AB	Illinois	Cancer	Succes
1	2024-08-02	Jennifer Brown	Female	68	B-	Illinois	Flu	Failure
2	2023-03-06	Jamie Lucas	Female	46	O	California	Infection	Succes
3	2024-11-09	Wayne Brown	Male	88	B-	Illinois	Injury	Failure
4	2023-12-20	Jennifer Fuller	Male	68	A+	Texas	Hypertension	Failure
...	...	...	...	...	...	...	...	..
1995	2024-02-02	Edward Roman	Male	29	AB	Florida	Diabetes	Succes
1996	2023-06-16	Barbara Shah Dvm	Male	65	B-	Illinois	Flu	Failure
1997	2022-08-09	Rebecca Davis	Male	10	B	New York	Flu	Succes
1998	2022-01-02	Stephen Daniels	Female	59	B	Illinois	Injury	Succes
1999	2021-08-30	Aaron Logan	Female	49	A	Texas	Hypertension	Succes

2000 rows × 9 columns

In [82]:

```
# Find rows where Patient Name does NOT match the expected format
df[~df["Patient Name"].str.match(r'^[A-Z][a-z]+ [A-Z][a-z]+$', na=False)]
```

Out[82]:

	Date of Admission	Patient Name	Gender	Age	Blood Type	Region	Medical Condition	Outcome
7	2021-12-01	Laura Clark Dds	Female	85	A	Florida	Diabetes	Failure
14	2022-10-10	Mrs. Julie Cohen Dvm	Male	16	AB	Florida	Cancer	Success
24	2023-01-15	Travis Beard Md	Female	15	B	Illinois	Hypertension	Success
41	2021-05-06	Catherine Watson Dds	Female	28	O	California	Diabetes	Success
42	2023-08-29	Ms. Tara Baker	Female	45	O	Illinois	Infection	Failure
...	...	...	...	...	...	...	...	...
1886	2023-06-23	Linda Crawford Dds	Male	14	B	Florida	Diabetes	Success
1913	2020-10-09	Cynthia Kaiser Dds	Female	44	B	California	Diabetes	Success
1928	2020-01-23	Jeffrey Allen Dvm	Male	46	A+	Florida	Cancer	Success
1940	2022-05-25	Mrs. Monica Williams	Male	69	B	Illinois	Cancer	Failure
1996	2023-06-16	Barbara Shah Dvm	Male	65	B-	Illinois	Flu	Failure

87 rows × 9 columns

In [83]:

```
# Remove titles (Mr, Mrs, Ms) from the beginning of Patient Name
df["Patient Name"] = df["Patient Name"].str.replace(
    r'^(Mr\.?|Mrs\.?|Ms\.?)\s+', '', regex=True
)

# Review the updated dataset
df
```

Out[83]:

	Date of Admission	Patient Name	Gender	Age	Blood Type	Region	Medical Condition	Outcome
0	2024-01-12	Kristopher Lopez	Female	47	AB	Illinois	Cancer	Succes
1	2024-08-02	Jennifer Brown	Female	68	B-	Illinois	Flu	Failure
2	2023-03-06	Jamie Lucas	Female	46	O	California	Infection	Succes
3	2024-11-09	Wayne Brown	Male	88	B-	Illinois	Injury	Failure
4	2023-12-20	Jennifer Fuller	Male	68	A+	Texas	Hypertension	Failure
...	...	...	...	...	...	...	...	..
1995	2024-02-02	Edward Roman	Male	29	AB	Florida	Diabetes	Succes
1996	2023-06-16	Barbara Shah Dvm	Male	65	B-	Illinois	Flu	Failure
1997	2022-08-09	Rebecca Davis	Male	10	B	New York	Flu	Succes
1998	2022-01-02	Stephen Daniels	Female	59	B	Illinois	Injury	Succes
1999	2021-08-30	Aaron Logan	Female	49	A	Texas	Hypertension	Succes

2000 rows × 9 columns

In [84]:

```
# Split names into parts
name_split = df["Patient Name"].str.split()

# Extract first name
df["First_Name"] = name_split.str[0]

# Extract last name
df["Last_Name"] = name_split.str[-1]

# Extract middle name(s) if present
df["Middle_Name"] = name_split.str[1:-1].apply(
    lambda x: ' '.join(x) if len(x) > 0 else ''
)

# Review the updated dataset
df
```

Out[84]:

	Date of Admission	Patient Name	Gender	Age	Blood Type	Region	Medical Condition	Outcome
0	2024-01-12	Kristopher Lopez	Female	47	AB	Illinois	Cancer	Succes
1	2024-08-02	Jennifer Brown	Female	68	B-	Illinois	Flu	Failure
2	2023-03-06	Jamie Lucas	Female	46	O	California	Infection	Succes
3	2024-11-09	Wayne Brown	Male	88	B-	Illinois	Injury	Failure
4	2023-12-20	Jennifer Fuller	Male	68	A+	Texas	Hypertension	Failure
...	...	...	...	...	...	...	...	..
1995	2024-02-02	Edward Roman	Male	29	AB	Florida	Diabetes	Succes
1996	2023-06-16	Barbara Shah Dvm	Male	65	B-	Illinois	Flu	Failure
1997	2022-08-09	Rebecca Davis	Male	10	B	New York	Flu	Succes
1998	2022-01-02	Stephen Daniels	Female	59	B	Illinois	Injury	Succes
1999	2021-08-30	Aaron Logan	Female	49	A	Texas	Hypertension	Succes

2000 rows × 12 columns

In [85]:
```
# Display summary information about the dataset
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Date of Admission  2000 non-null   datetime64[us]
 1   Patient Name       2000 non-null   str
 2   Gender             2000 non-null   str
 3   Age                2000 non-null   int64
 4   Blood Type         2000 non-null   str
 5   Region             2000 non-null   str
 6   Medical Condition  2000 non-null   str
 7   Outcome            2000 non-null   str
 8   Date of Discharge  2000 non-null   datetime64[us]
 9   First_Name         2000 non-null   object
 10  Last_Name          2000 non-null   object
 11  Middle_Name        2000 non-null   str
dtypes: datetime64[us](2), int64(1), object(2), str(7)
memory usage: 187.6+ KB
```

In [86]:
```
# Convert selected columns from object to string type
cols = ["Gender", "Blood Type", "Region",
        "Medical Condition", "Outcome",
```

```
        "First_Name", "Last_Name", "Middle_Name"]

df[cols] = df[cols].astype("string")

# Review the updated dataset
df
```

Out[86]:

	Date of Admission	Patient Name	Gender	Age	Blood Type	Region	Medical Condition	Outcome
0	2024-01-12	Kristopher Lopez	Female	47	AB	Illinois	Cancer	Succes
1	2024-08-02	Jennifer Brown	Female	68	B-	Illinois	Flu	Failure
2	2023-03-06	Jamie Lucas	Female	46	O	California	Infection	Succes
3	2024-11-09	Wayne Brown	Male	88	B-	Illinois	Injury	Failure
4	2023-12-20	Jennifer Fuller	Male	68	A+	Texas	Hypertension	Failure
...	...	...	...	...	...	...	...	..
1995	2024-02-02	Edward Roman	Male	29	AB	Florida	Diabetes	Succes
1996	2023-06-16	Barbara Shah Dvm	Male	65	B-	Illinois	Flu	Failure
1997	2022-08-09	Rebecca Davis	Male	10	B	New York	Flu	Succes
1998	2022-01-02	Stephen Daniels	Female	59	B	Illinois	Injury	Succes
1999	2021-08-30	Aaron Logan	Female	49	A	Texas	Hypertension	Succes

2000 rows × 12 columns

In [87]:

```
# Check the data type
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Date of Admission  2000 non-null   datetime64[us]
 1   Patient Name       2000 non-null   str
 2   Gender             2000 non-null   string
 3   Age                2000 non-null   int64
 4   Blood Type         2000 non-null   string
 5   Region             2000 non-null   string
 6   Medical Condition  2000 non-null   string
 7   Outcome            2000 non-null   string
 8   Date of Discharge  2000 non-null   datetime64[us]
 9   First_Name         2000 non-null   string
 10  Last_Name          2000 non-null   string
 11  Middle_Name        2000 non-null   string
dtypes: datetime64[us](2), int64(1), str(1), string(8)
memory usage: 187.6 KB
```

In [88]:
```
# Check unique values in Gender column
df["Gender"].unique()
```

Out[88]:
```
<StringArray>
['Female', 'Male']
Length: 2, dtype: string
```

In [89]:
```
# Check unique values in Age column
df["Age"].describe()
```

Out[89]:
```
count    2000.000000
mean       50.280000
std        23.578647
min        10.000000
25%        30.000000
50%        51.000000
75%        71.000000
max        90.000000
Name: Age, dtype: float64
```

In [90]:
```
# Check unique values in Blood Type column
df["Blood Type"].unique()
```

Out[90]:
```
<StringArray>
['AB', 'B-', 'O', 'A+', 'A', 'B']
Length: 6, dtype: string
```

In [91]:
```
# Check unique values in Region column
df["Region"].unique()
```

Out[91]:
```
<StringArray>
['Illinois', 'California', 'Texas', 'New York', 'Florida']
Length: 5, dtype: string
```

In [92]:
```
# Check unique values in Medical Condition column
df["Medical Condition"].unique()
```

Out[92]:
```
<StringArray>
['Cancer', 'Flu', 'Infection', 'Injury', 'Hypertension', 'Diabetes']
Length: 6, dtype: string
```

In [93]:
```
# Check unique values in Outcome column
df["Outcome"].unique()
```

Out[93]:
```
<StringArray>
['Success', 'Failure']
Length: 2, dtype: string
```

In [ ]:

In [94]:
```
# Check the data type of the Date of Discharge column
df["Date of Discharge"].dtype
```

Out[94]:
```
dtype('<M8[us]')
```

In [95]:
```
# Check if there are any missing values in the dataset
df.isna().any().any()
```

Out[95]:
```
np.False_
```

In [96]:
```
# Remove Patient Name column as it is not needed for analysis
# df = df.drop(columns=["Patient Name"])
df
```

Out[96]:

	Date of Admission	Patient Name	Gender	Age	Blood Type	Region	Medical Condition	Outcome
0	2024-01-12	Kristopher Lopez	Female	47	AB	Illinois	Cancer	Succes
1	2024-08-02	Jennifer Brown	Female	68	B-	Illinois	Flu	Failure
2	2023-03-06	Jamie Lucas	Female	46	O	California	Infection	Succes
3	2024-11-09	Wayne Brown	Male	88	B-	Illinois	Injury	Failure
4	2023-12-20	Jennifer Fuller	Male	68	A+	Texas	Hypertension	Failure
...	...	...	...	...	...	...	...	..
1995	2024-02-02	Edward Roman	Male	29	AB	Florida	Diabetes	Succes
1996	2023-06-16	Barbara Shah Dvm	Male	65	B-	Illinois	Flu	Failure
1997	2022-08-09	Rebecca Davis	Male	10	B	New York	Flu	Succes
1998	2022-01-02	Stephen Daniels	Female	59	B	Illinois	Injury	Succes
1999	2021-08-30	Aaron Logan	Female	49	A	Texas	Hypertension	Succes

2000 rows × 12 columns

In [97]:
```
# Export cleaned dataset to Excel file
df.to_excel("cleaned_data.xlsx", index=False)
```

In [ ]: